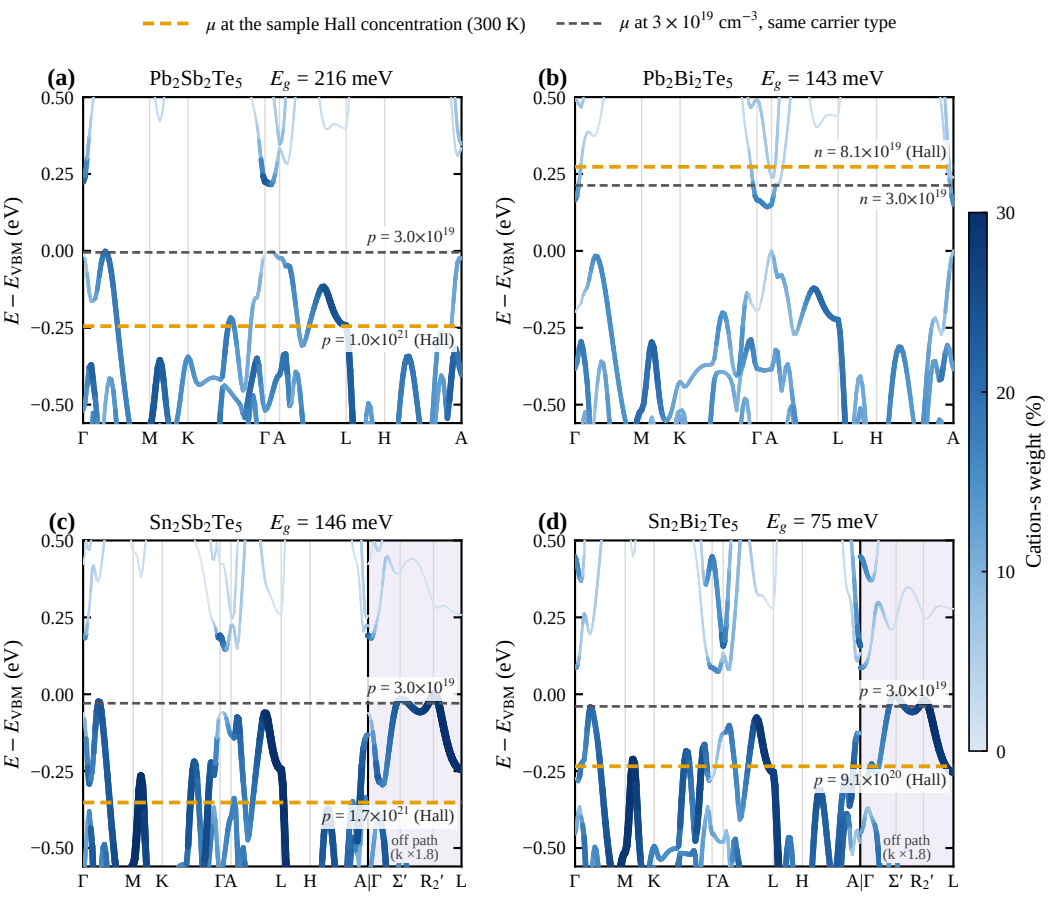

μ at the sample Hall concentration (300 K)
μ at 3 × 10^19 cm^−3, same carrier type
(a)
Pb2Sb2Te5 Eg = 216 meV
p = 3.0×10^19
p = 1.0×10^21 (Hall)
(b)
Pb2Bi2Te5 Eg = 143 meV
n = 8.1×10^19 (Hall)
n = 3.0×10^19
(c)
Sn2Sb2Te5 Eg = 146 meV
p = 3.0×10^19
p = 1.7×10^21 (Hall)
off path (k ×1.8)
(d)
Sn2Bi2Te5 Eg = 75 meV
p = 3.0×10^19
p = 9.1×10^20 (Hall)
off path (k ×1.8)
E − EVBM (eV)
0.50
0.25
0.00
−0.25
−0.50
Γ M K ΓA L H A
Γ M K ΓA L H A|Γ Σ′ R2′ L
Cation-s weight (%)
30
20
10
0